

RelicVault AI — Project Write-up Summary

A crowdsourced digital heritage & minor-artifact museum · Next.js 14 · MongoDB · OpenAI Vision · OpenStreetMap · 3-language UI

1 person 1 summer Full-stack · AI-powered Multi-language Privacy-by-default

Problem statement

China has a long history and a vast amount of cultural heritage, yet many artifacts are not properly preserved, and many valuable ones remain little known. Amateur finds — village heirlooms, family keepsakes, flea-market objects — have no lightweight, AI-assisted home, so they stay unprotected and forgotten rather than becoming part of a shared archive.

Solution overview

RelicVault AI is a full-stack web app. Users sign up, drop a photo of an artifact plus a map pin, and instantly get a structured record (name, dynasty, category, material, preservation, tags) from a vision LLM. Explore offers fuzzy search, `#tag` search, era / material / preservation filters, and a grid-or-map toggle. Coordinates blur to two decimals before storage.

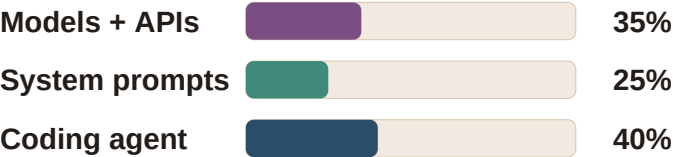
Use of AI

AI played three roles this summer. The breakdown is in the bars below; the details live in the Reflections strip.

1 · Coding agent (WorkBuddy). scaffolds, refactors, CSS-animation debugging.

2 · System prompts. strict JSON schema + 'say unknown' + bilingual reasoning.

3 · Models & APIs. OpenAI vision, OpenStreetMap Nominatim, Leaflet.



GPS pin = OSM Leaflet picker; coordinates blurred to 2 decimals before storage.

Reflections

What worked

- Started with the data model. The Mongoose schema came first; once right, every UI feature became a thin layer on top.
- Treated the prompt as a product spec — JSON schema + 'say unknown' = auditable vision pipeline.
- Small dependency surface: one LLM, one geocoder, one map library, one database.

Challenges encountered

- CSS entrance animations — the AI agent performed poorly here. `@keyframes` stripped inside `@layer`; `display:contents` killing transforms. Three hard rules now live in my project memory.
- Permissions gap on the artifacts collection (writes vs. reads) — solved at the application layer.

Future enhancements

- Xiaohongshu-style social layer — public profiles, follow graph, direct messages.
- Richer uploads — multiple images, short video, and pure-text entries.
- Stronger auth — third-party sign-in (Google / WeChat), email & phone-number verification.

Impact & value. For amateur collectors and local history enthusiasts, the product lowers the cost of contributing to a shared cultural archive from days of research to under a minute; coordinates are blurred by default so dig-site secrecy is preserved. For researchers & educators, the structured JSON output is directly queryable, exportable, and mappable.